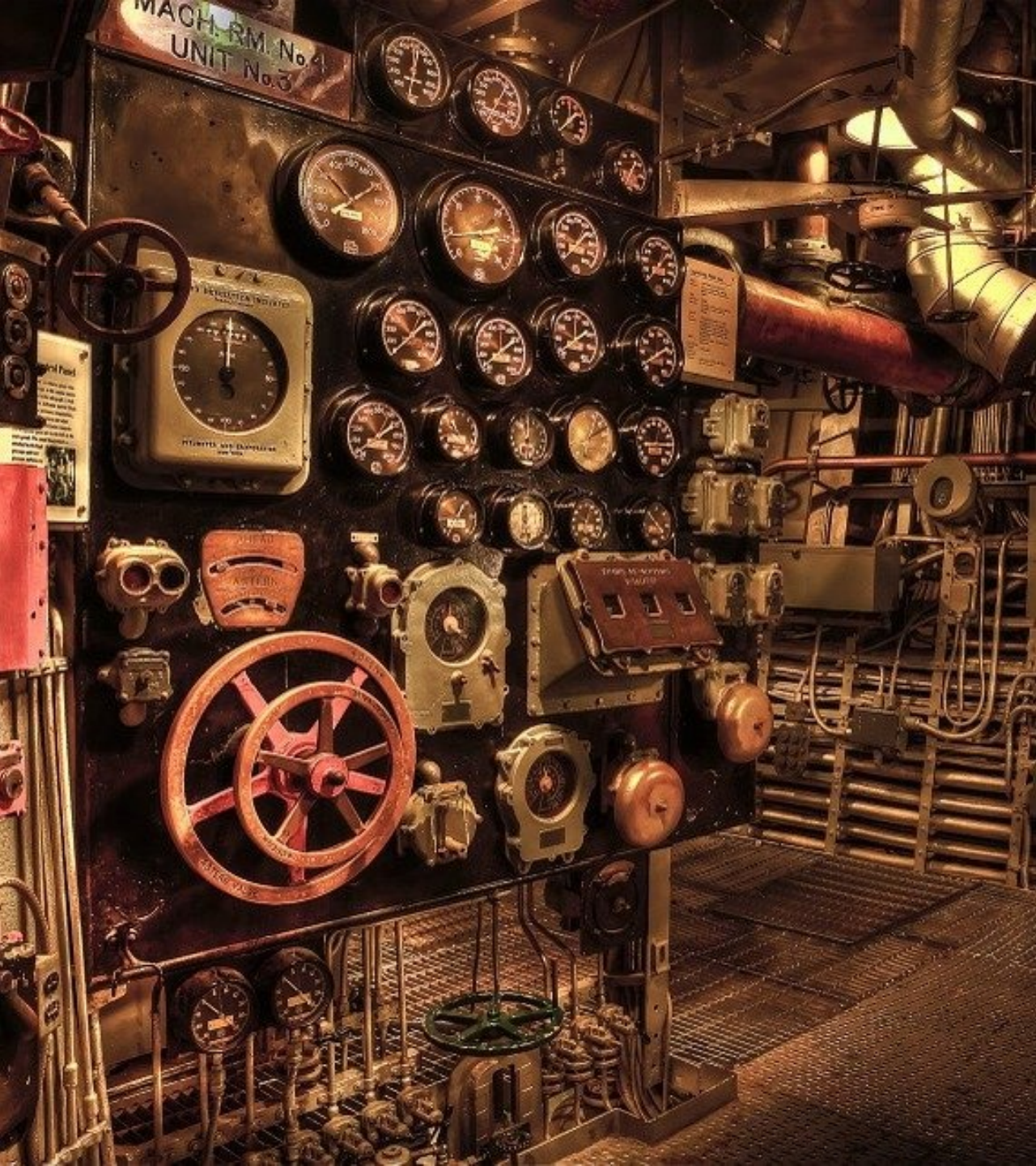




ME 300: Engineering Thermodynamics

Carnot Efficiency

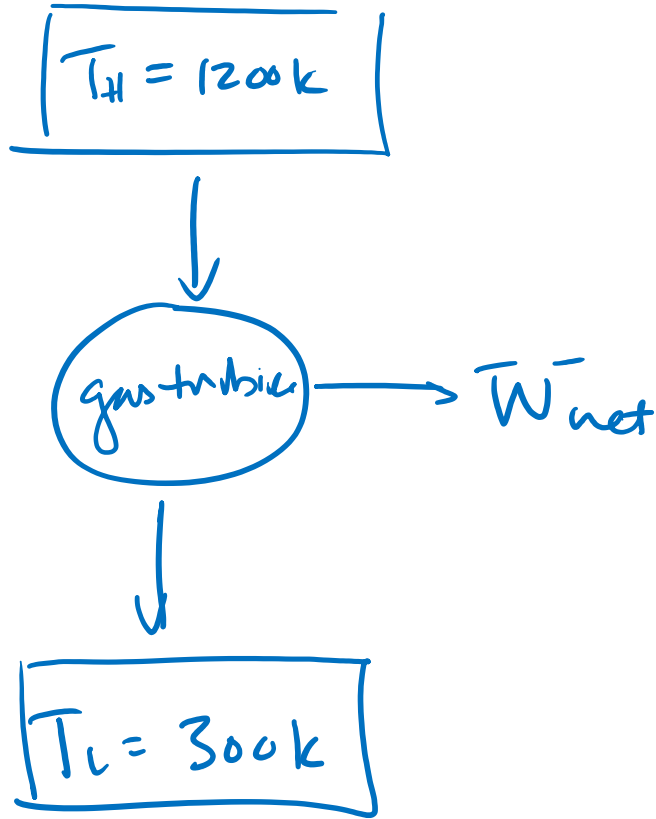
Carnot efficiency

$$\eta_{th, rev} = \underbrace{\eta_{th, max}}_{\text{Carnot efficiency}} = 1 - \frac{T_L}{T_H} \quad \rightarrow \text{absolute temp scale - kelvin}$$

$$\hookrightarrow \eta_{th} \neq 1, \eta_{th} < 1 \rightarrow \begin{matrix} T_L > 0 \\ T_H < \infty \end{matrix}$$

Example: Carnot efficiency

What is the greatest efficiency possible for a gas turbine where heat is added at 1200 K and the gas rejects heat to the atmosphere at 300 K?



$$\eta_{th, max} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{1200}$$

$$\eta_{th, max} = 0.75$$

Carnot cycle - ideal, reversible

Process 1-2: isothermal heat addition

2-3: adiabatic expansion

3-4: isothermal heat rejection

4-1: adiabatic compression.